

★ WEEK 3 · PART 2 of 3

ARMA and Identification

Combining the Engines, and the Box-Jenkins Method

ARMA(p,q) · the ACF/PACF cheat sheet · the identify–estimate–diagnose loop

Part 1 gave us AR and MA separately. Part 2 fuses them into ARMA and turns identification into a disciplined procedure — the Box-Jenkins method that made time-series modeling teachable.

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PART 2 · IDENTIFICATION

Choosing p and q , With Discipline

ARMA in one framework, the ACF/PACF signatures that reveal the order, and the iterative Box-Jenkins loop from raw data to a validated model.

From Two Engines to One Method

Part 1 gave the parts; Part 2 assembles them

1

We can now read AR and MA fingerprints

PACF cut-off means AR; ACF cut-off means MA. That single reading is the foundation everything here builds on.

2

Real series usually need both

Combining feedback (AR) and shock echoes (MA) yields ARMA — often a far more compact fit than either alone.

3

Identification becomes a procedure

Box and Jenkins turned 'which model?' into a repeatable loop: identify, estimate, diagnose, and repeat until the residuals are clean.

Part 2 in one line: combine AR and MA, then use the ACF/PACF to choose p and q .

ARMA(p,q): Combining Both

Feedback and shock echoes in one model

$$X_t = c + \sum_{i=1}^p \phi_i X_{t-i} + \varepsilon_t + \sum_{j=1}^q \theta_j \varepsilon_{t-j}$$

The two halves

- The AR part: today depends on its own past p values.
- The MA part: today also echoes the last q shocks.
- Together they describe both persistence and short-lived surprises.

Reading (p, q)

- p is the autoregressive order; q is the moving-average order.
- ARMA(1,0) is AR(1); ARMA(0,1) is MA(1).
- Most well-behaved series need only small p and q .

ARMA in Lag-Operator Form

One compact line captures the whole family

$$\phi(L) X_t = c + \theta(L) \varepsilon_t \quad (\text{ARMA in lag-operator form})$$

Why this form

- $\phi(L)$ is the AR polynomial; $\theta(L)$ is the MA polynomial.
- The whole model is one polynomial equation in L .
- Stationarity and invertibility are just root conditions on each.

What it unifies

- AR is $\theta(L) = 1$; MA is $\phi(L) = 1$; ARMA has both non-trivial.
- It makes the algebra of forecasting mechanical.
- Every stats package works in exactly this representation.

Why Combine? Parsimony

A small ARMA can replace a large pure model

A long-memory series might need an AR(10) — or an ARMA(1,1) with just two parameters. Combining the engines often buys the same fit with far fewer coefficients.

The trade you want

- Fewer parameters mean less to estimate and less to overfit.
- A tight ARMA generalizes better than a bloated AR or MA.
- This is the whole reason the ARMA class exists.

The cost

- ARMA is harder to identify — both plots tail off.
- So identification takes more judgment than pure AR or MA.
- The cheat sheet and the loop are how we manage that.

Choosing p and q

Let the data's autocorrelations guide you

Identification means reading the sample ACF and PACF to propose a candidate (p, q) . It is the first — and most skill-dependent — step of the whole method.

What we look for

- Where does the ACF cut off, tail off, or die slowly?
- Where does the PACF cut off or tail off?
- The pattern points to AR, MA, or ARMA — and its order.

Why it is hard

- Sample plots are noisy — no cut-off is perfectly sharp.
- ARMA makes both plots tail off, hiding the order.
- So we propose, fit, and check — never guess once.

The Identification Cheat Sheet

Memorize this — you will use it constantly

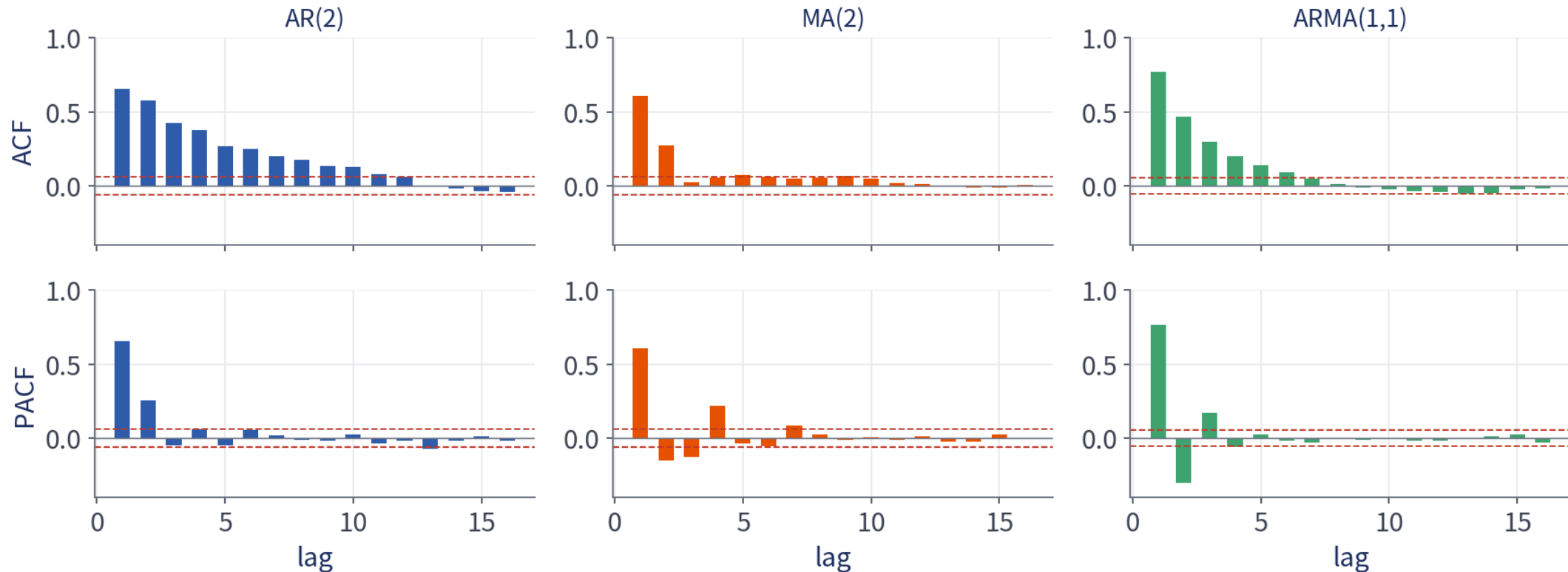
Model	ACF (total)	PACF (direct)
AR(p)	tails off (geometric / damped)	cuts off sharply after lag p
MA(q)	cuts off sharply after lag q	tails off (geometric / damped)
ARMA(p,q)	tails off	tails off

The rule: the plot that CUTS OFF names the order. If BOTH tail off, you are looking at an ARMA.

The Cheat Sheet, Visualized

AR(2) cuts its PACF; MA(2) cuts its ACF; ARMA(1,1) tails off in BOTH — the tell-tale that neither pure model will do.

Signature gallery: AR cuts PACF, MA cuts ACF, ARMA tails off in BOTH



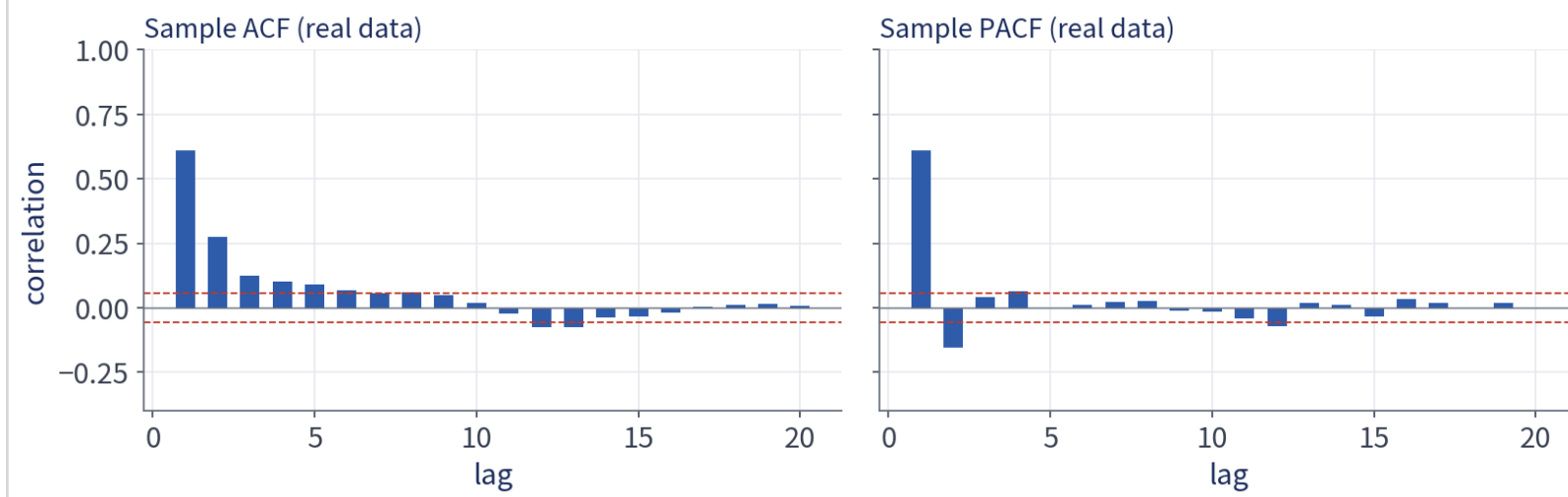
Real Data Is Never So Tidy

On real series, sampling noise blurs the cut-offs — identification becomes a judgment call, not a lookup.

What to expect

- No razor-sharp cut-off.
- Bars hover near the band.
- Several (p,q) look plausible.
- So propose a few.
- Then let diagnostics decide.

Real data is noisy: no clean cut-off — identification is a judgment call



The Box-Jenkins Methodology

A disciplined loop, not a one-shot recipe

Box-Jenkins replaces guesswork with a cycle: identify a candidate model, estimate it, check its residuals, and repeat until nothing systematic is left behind.

Why a loop

- The first candidate is rarely the final model.
- Diagnostics reveal what identification missed.
- Each pass refines p and q toward a clean fit.

The stopping rule

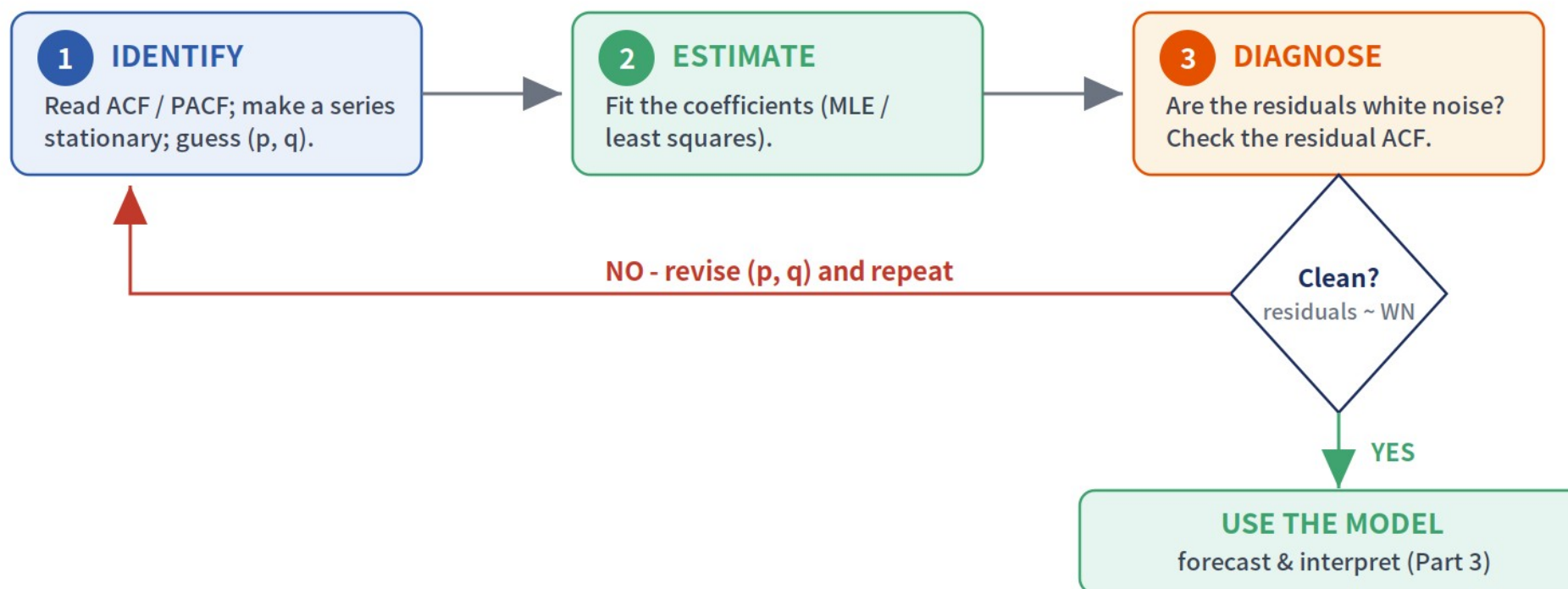
- Stop when the residuals look like white noise.
- At that point the model has captured all the structure.
- Then — and only then — do you forecast.

Box-Jenkins: The Iterative Loop

Identify, estimate, diagnose — and if the residuals aren't clean, revise (p, q) and go around again. Only white-noise residuals let you exit to forecasting.

The Box-Jenkins Loop

Not a one-shot recipe - a disciplined cycle you repeat until the residuals are clean.



Identification in Practice

Turning plots into a candidate model

The routine

- Make the series stationary first (Week 2): difference or detrend.
- Plot the sample ACF and PACF out to a couple dozen lags.
- Match the pattern to the cheat sheet and propose (p, q) .

Practical tips

- Keep p and q small to start — parsimony first.
- When both plots tail off, try a low-order ARMA like $(1,1)$.
- Note two or three candidates; you will compare them later.

Estimation in Brief

How the coefficients are actually fit

$$\hat{\theta} = \arg \min_{\theta} \sum_t \hat{\varepsilon}_t^2 \quad \text{or} \quad \arg \max_{\theta} \mathcal{L}(\theta)$$

The idea

- Choose coefficients that make the model's errors as small as possible.
- Equivalently, maximize the likelihood of the observed data.
- The software solves this numerically — you read the output.

What you get back

- Estimated ϕ 's and θ 's, each with a standard error.
- A log-likelihood you will reuse for AIC and BIC.
- Residuals to feed into the diagnostic step.

Estimation, Intuitively

What the optimizer is really doing

The fit hunts for the coefficients under which the model would have predicted the past best — the ones that leave the smallest, most structureless residuals.

Least squares view

- Every candidate model implies one-step-ahead predictions.
- Sum the squared prediction errors across the sample.
- The best coefficients minimize that sum.

Likelihood view

- Assume the shocks are (roughly) normal.
- Pick coefficients that make the data most probable.
- For Gaussian shocks the two views nearly coincide.

Diagnostic Checking

The gate you must pass before trusting a model

A good ARMA leaves residuals that are indistinguishable from white noise. If the residual ACF still shows structure, the model has missed something — go back and revise.

What to check

- Plot the residual ACF — no spikes should stand out.
- Run a portmanteau test (e.g. Ljung-Box) on the residuals.
- Look for remaining patterns, outliers, or changing variance.

How to respond

- A leftover spike at lag k suggests raising p or q .
- Changing variance points to a volatility model (later weeks).
- Iterate until the residuals are clean — that is the exit.

Parsimony: The Guiding Principle

The simplest model that works, wins

Between two models that fit equally well, always prefer the one with fewer parameters. Extra coefficients fit noise, not signal — and noise does not repeat out of sample.

Why simpler is safer

- Fewer parameters mean smaller estimation error.
- Simple models generalize; complex ones memorize the sample.
- This is the bias-variance trade-off in one sentence.

How we enforce it

- Start small and add order only when diagnostics demand it.
- Penalize complexity explicitly with AIC and BIC (next).
- Box's warning again: a useful model, not a perfect one.

Common ARMA Specifications

A short menu that covers most cases

Model	What it captures	Typical use
AR(1)	simple persistence / mean reversion	interest rates, spreads, ratios
MA(1)	a one-period shock echo	microstructure noise, revisions
ARMA(1,1)	persistence plus a short echo	many macro & financial series
AR(2)	cycles and overshooting	business-cycle-like dynamics

Start with one of these; escalate order only if the residuals still show structure.

Model Selection: AIC and BIC

Balancing fit against complexity

$$\text{AIC} = -2 \ln \hat{\mathcal{L}} + 2k, \quad \text{BIC} = -2 \ln \hat{\mathcal{L}} + k \ln T$$

How to read them

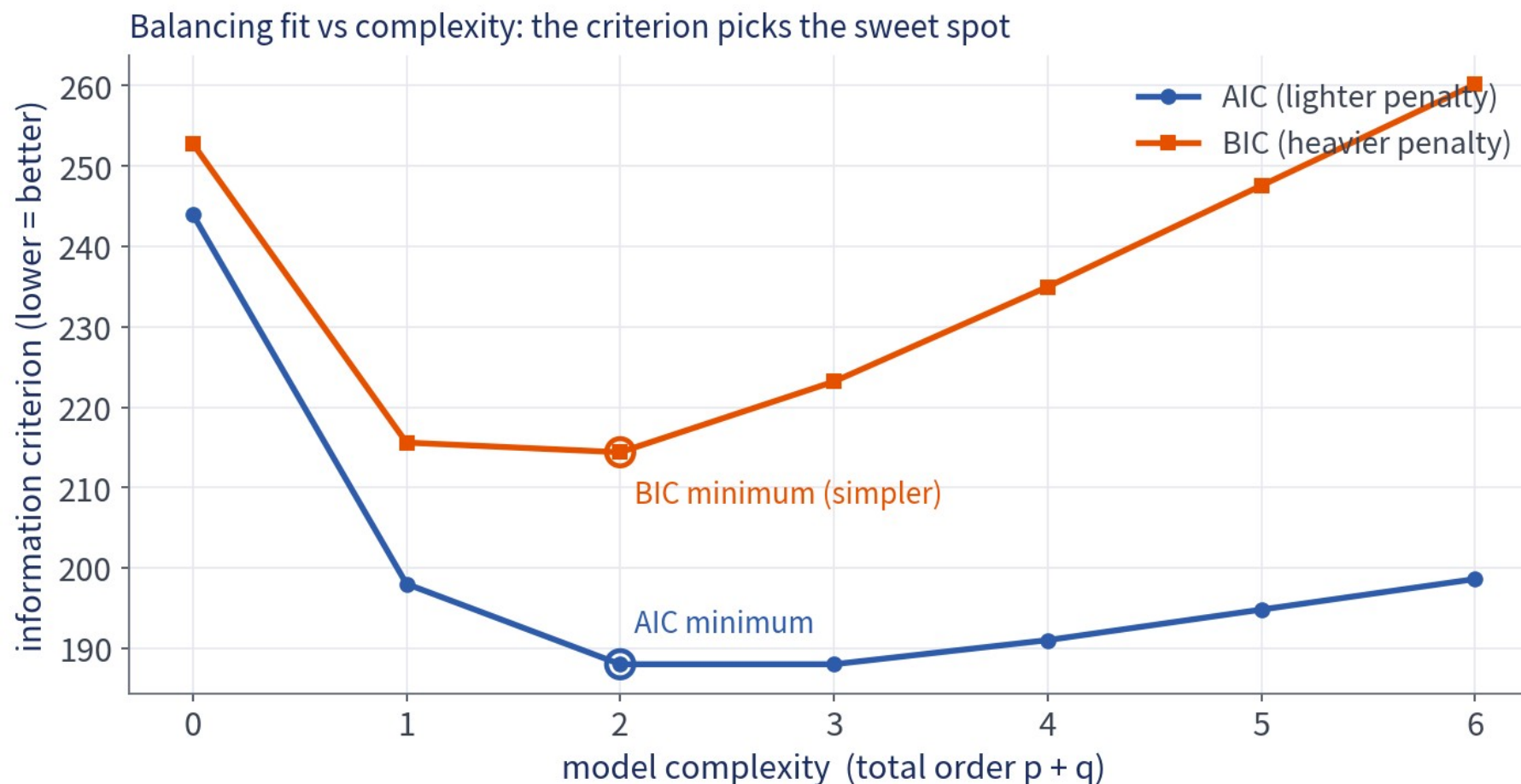
- Both reward fit (the likelihood) and penalize the parameter count k .
- Lower is better; compare candidates and pick the minimum.
- BIC's penalty ($k \cdot \ln T$) is heavier, so it favors simpler models.

How to use them

- Fit a small grid of (p, q) candidates.
- Rank by AIC or BIC and shortlist the best.
- Then confirm the winner with residual diagnostics.

Fit vs Complexity, Made Visible

As order grows, fit improves but the penalty rises — the criterion dips to a minimum. BIC's heavier penalty lands on a simpler model than AIC.



Overfitting: The Ever-Present Danger

When more fit means worse forecasts

Add enough parameters and any model fits the past perfectly — including its noise. That noise will not recur, so the over-fit model forecasts worse, not better.

The symptoms

- In-sample fit keeps improving as you add order.
- But out-of-sample accuracy stalls, then degrades.
- Coefficients become large, unstable, or barely significant.

The safeguards

- Let AIC/BIC penalize complexity for you.
- Validate on held-out data, not just the training sample.
- When in doubt, choose the simpler model.

Automate, Then Judge

How Box-Jenkins is run today

What the tools do

- Automated search (e.g. `auto_arima`) grids over (p, q) by AIC/BIC.
- It returns a ranked shortlist in seconds.
- This handles the tedious search you once did by hand.

What stays your job

- Making the series stationary correctly (Week 2).
- Reading the residual diagnostics and sanity-checking the pick.
- Judging whether the model makes economic sense.

Box-Jenkins, End to End

Direct Claude Code through the full identify–estimate–diagnose loop

GOAL Run the complete Box-Jenkins loop on a simulated ARMA series, then confirm you recovered the right order.

[INPUT] Prompt · Claude Code

Write and run `box_jenkins.py` that:

1. simulates a known ARMA(1,1) (hide the parameters from the printed output);
2. plots its ACF and PACF and proposes a couple of candidate (p, q) orders;
3. fits each candidate, reports AIC and BIC, and picks the best;
4. runs a Ljung-Box test on the residuals and plots their ACF.

Finally, reveal the true order and tell me whether the loop recovered it.

How to work with it

- Expect ARMA(1,1) to win on AIC/BIC.
- Residual ACF should look like white noise.
- Try an under-fit AR(1) — see the leftover spike.
- Keep it; Part 3 adds forecasting on top.

Deliverable: `box_jenkins.py` + the ACF/PACF and residual plots for one series.

Three Questions Before You Move On

If you can answer these, Part 2 has landed

Q1

Both the ACF and PACF tail off gradually. What model class does this suggest?

ARMA(p,q)

Q2

AIC picks ARMA(2,2); BIC picks ARMA(1,1). Which is likely simpler, and why?

BIC — heavier penalty

Q3

Your fitted model's residual ACF still has a spike at lag 1. Next step?

Revise p or q

All three come straight from the cheat sheet, the criteria, and the diagnostic step.

Three Pitfalls to Avoid

Where identification goes wrong

Common mistakes

- "Identify once and trust it."
- "The lowest AIC is always the right model."
- "Skip stationarity and go straight to ARMA."

The reality

- Box-Jenkins is a loop — diagnostics can send you back to step 1.
- AIC can overfit; cross-check with BIC and residual diagnostics.
- ARMA assumes stationarity — difference or detrend first (Week 2).

Three Things to Carry Forward

Part 2 in one screen

1

ARMA fuses feedback and echoes — parsimoniously

A small ARMA(p,q) often matches what a large pure AR or MA would need, capturing persistence and shocks in one compact model.

2

The ACF/PACF cheat sheet identifies the order

A cut-off names the order (ACF→MA, PACF→AR); both tailing off means ARMA. On real data, propose a few candidates and compare.

3

Box-Jenkins is a disciplined loop

Identify, estimate, diagnose — guided by AIC/BIC and parsimony — and repeat until the residuals are white noise.

Next — Part 3: diagnostics in depth, forecasting, and a humbling truth about returns.